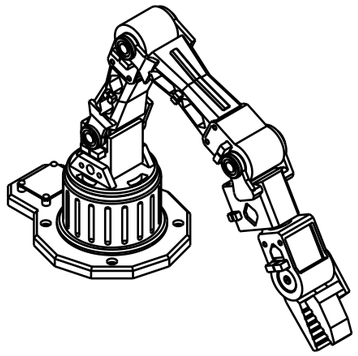
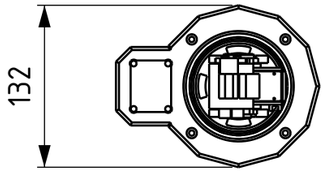
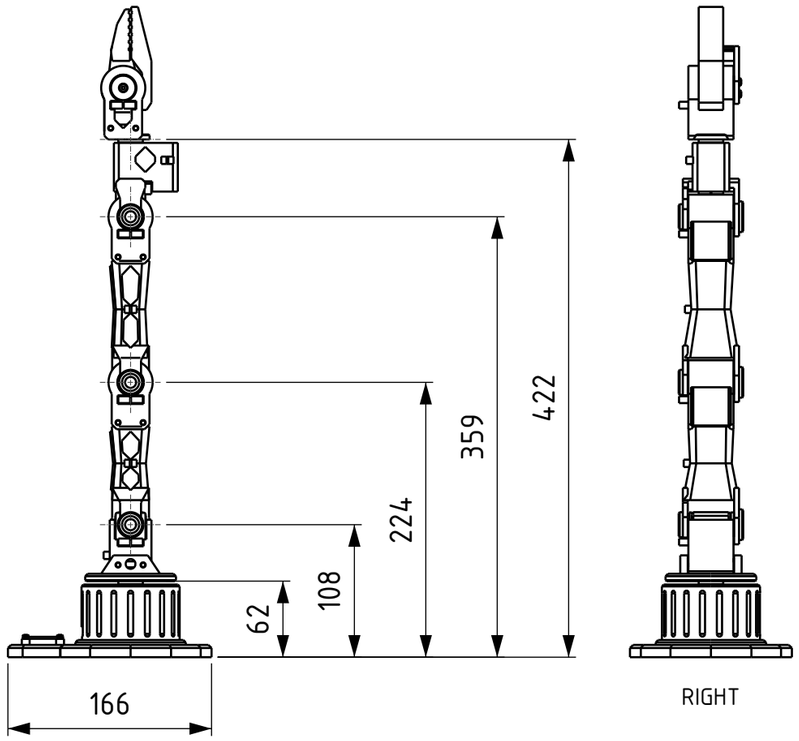


ISOMETRIC (NOT TO SCALE)



TOP



FRONT

RIGHT

PRINTED PARTS

PART NO.	NAME	QTY	SET
TAL-101	Base	1	A
TAL-102	Shoulder (turntable + cradle)	1	A
TAL-103	Upper arm	1	A
TAL-104	Forearm	1	B
TAL-105	Wrist	1	B
TAL-106	Gripper body	1	B
TAL-107	Moving jaw	1	B
TAL-108	Hub cap, large	3	A
TAL-109	Hub cap, small	3	A

BOUGHT PARTS

ITEM	QTY
Feetech STS3215 servo (12 V)	6
Waveshare Bus Servo Adapter (A)	1
M3 x 6 button head (servo horns)	24
M3 x 8 button head (passive horns)	3
M2 x 5 self-tapping (servo cases)	12
M2 x 6 self-tapping (servo cases)	8
M2.5 x 6 self-tapping (board)	4
M4 bolts to the table (user's)	4
3-pin bus servo cables (with servos)	6

JOINT LAYOUT

1. Pan axis vertical through the base centre; pan horn face 62 above the table. Plinth 132 across corners.
2. Shoulder axis (along Y) 46 above the pan horn face. Upper arm 116, forearm 135 (axis to axis). Wrist flex axis to roll horn face 63.
3. Front view: zero pose (all joints 0). Isometric: 'ready' pose.

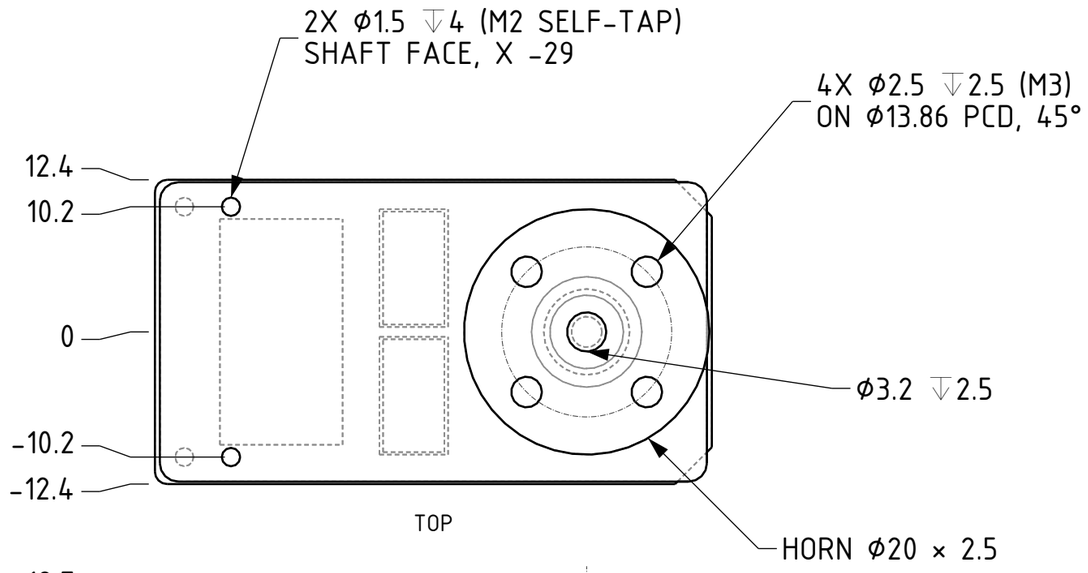
NOTES

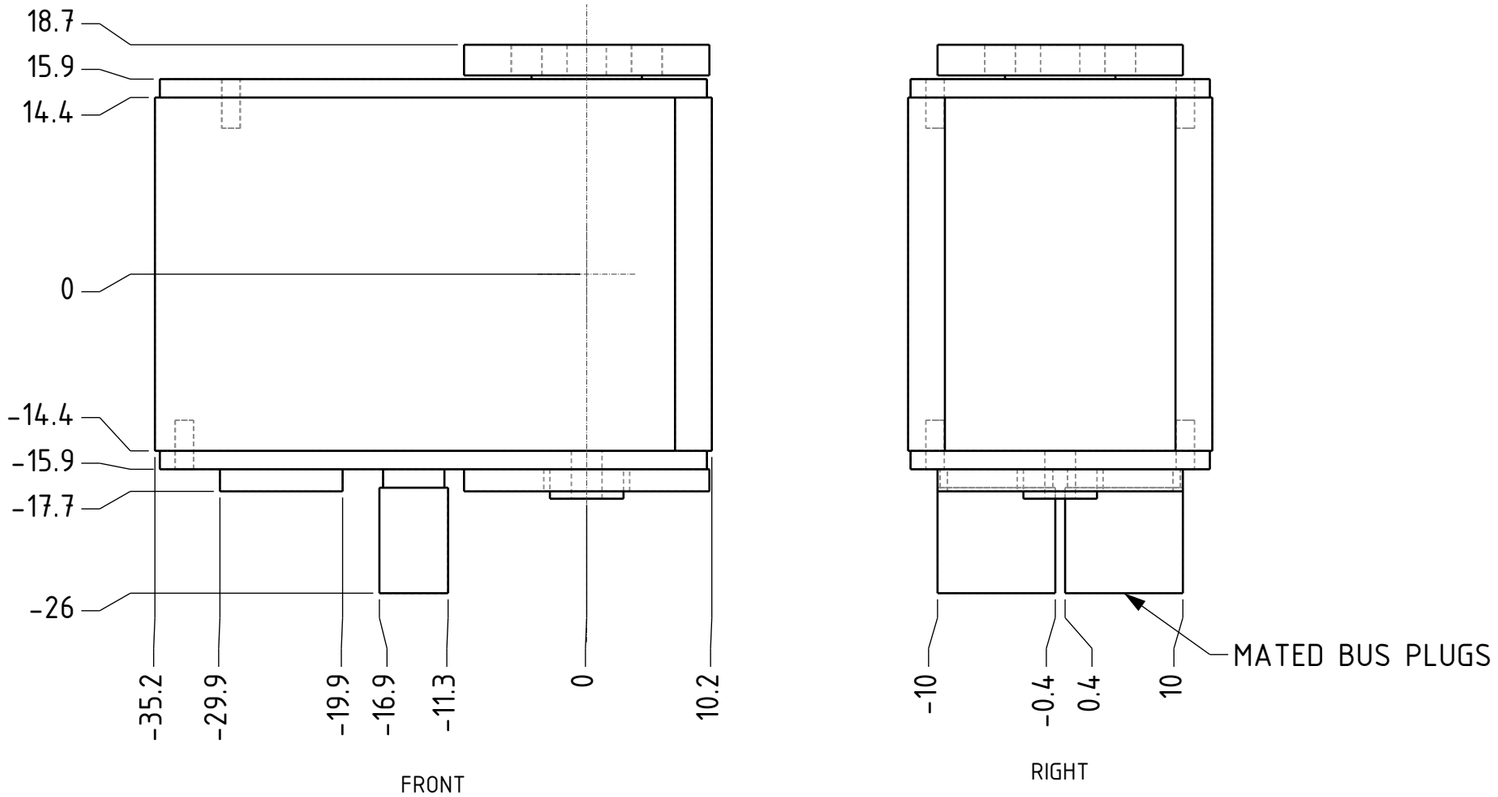
- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part sheets give each part in its own frame; this sheet shows where those frames sit.

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B

PROJECT TALOS 5-axis arm	EST. MASS -
PART General assembly and parts list	SCALE 1:5
PART NO. TAL-000	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-04
MATERIAL -	SHEET 1 of 20





SERVO (servo frame)

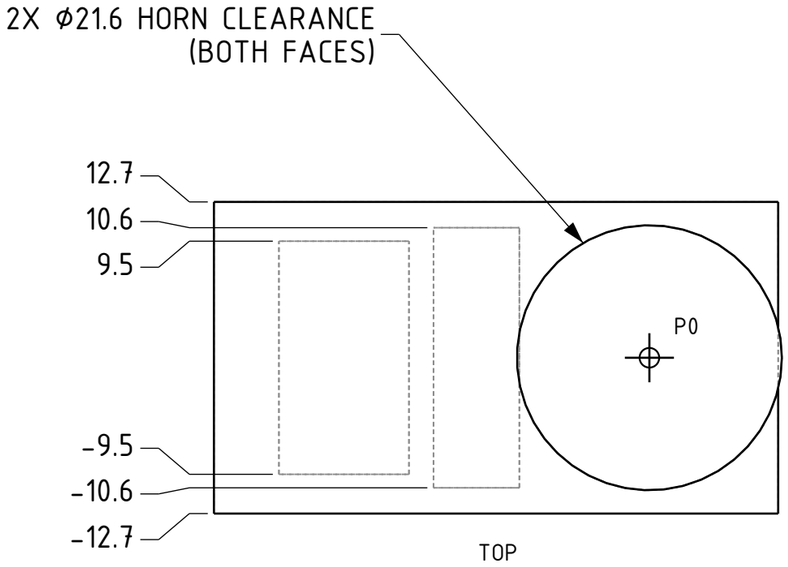
1. Body 45.4 × 24.8 × 28.8, x −35.2 to 10.2; +X end corners C3, −X end corners R1.
2. Caps 44.6 × 24.4 × 1.5, corners R1.5, centred on x −12.5, on both faces.
3. Back pad (connector face) x −29.9 to −19.9, y ±9.2, 1.8 high.
4. Bus headers x −16.6 to −11.6, y 0.6 to 9.8 and −9.8 to −0.6; mated plugs to z −26.
5. Output horn ⌀20 × 2.5 (z 16.2 to 18.7): 4X ⌀2.5 M3 on ⌀13.86 PCD at 45°, centre ⌀3.2.
6. Passive horn ⌀20 × 1.8 on the connector face, centre ⌀7.
7. Case holes ⌀1.5 ↧4 (M2 self-tap): shaft face at (−29, ±10.2), connector face at (−32.8, ±10.2).

NOTES

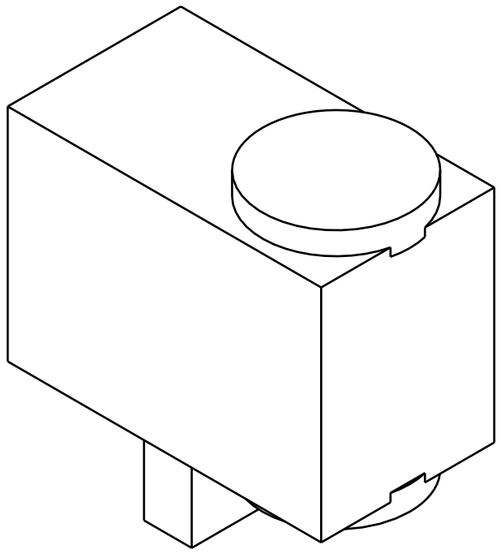
- Bought part: dimensions for reference, measured from the SO-ARM100 STS3215 model and Feetech's drawing. Check the connector position and case-hole depth on a real servo.
- Sheet 2 gives pocket P, the clearance envelope that every servo part cuts.

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B	
PROJECT TALOS 5-axis arm	EST. MASS -
PART STS3215 servo (reference) (part sheet 1 of 2)	SCALE 2:1
PART NO. TAL-001	UNITS / REV mm / A
QTY / ARM 6	DATE 2026-11-04
MATERIAL bought part	SHEET 2 of 20



TOP



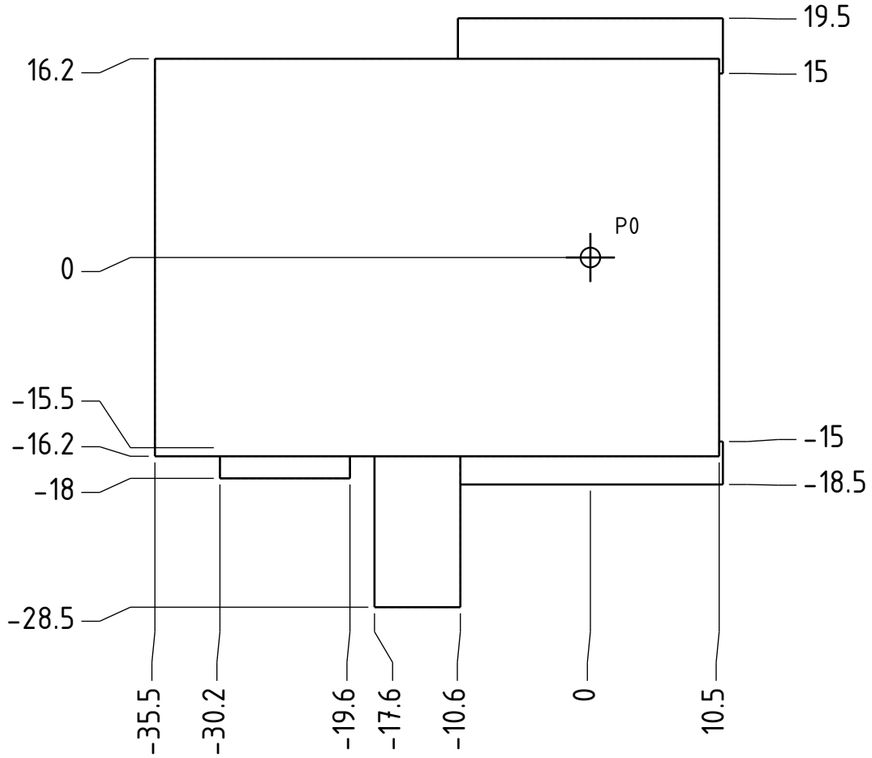
ISOMETRIC (NOT TO SCALE)

POCKET P (servo frame; 0.3 clearance included)

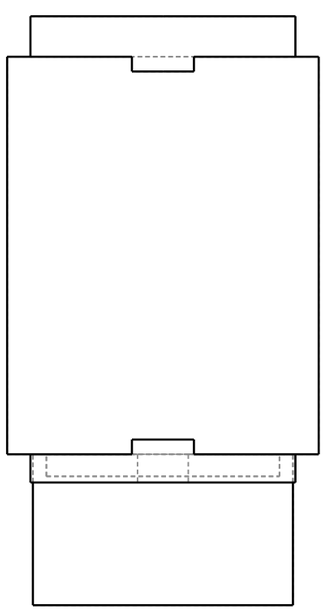
1. Main box x −35.5 to 10.5, y ±12.7, z ±16.2.
2. Back pad box x −30.2 to −19.6, y ±9.5, z −18.0 to −15.5.
3. Plug bay box x −17.6 to −10.6, y ±10.6, z −28.5 to −15.5 (mated plugs + cable bend).
4. Idler clearance cylinder ⌀21.6 on the shaft axis, z −18.5 to −15.0.
5. Horn clearance cylinder ⌀21.6 on the shaft axis, z 15.0 to 19.5.

NOTES

- Each part sheet locates P0 and gives the servo axes in the part frame; cut the union of the five bodies.
- All dimensions mm. Third-angle projection. Do not scale the drawing.



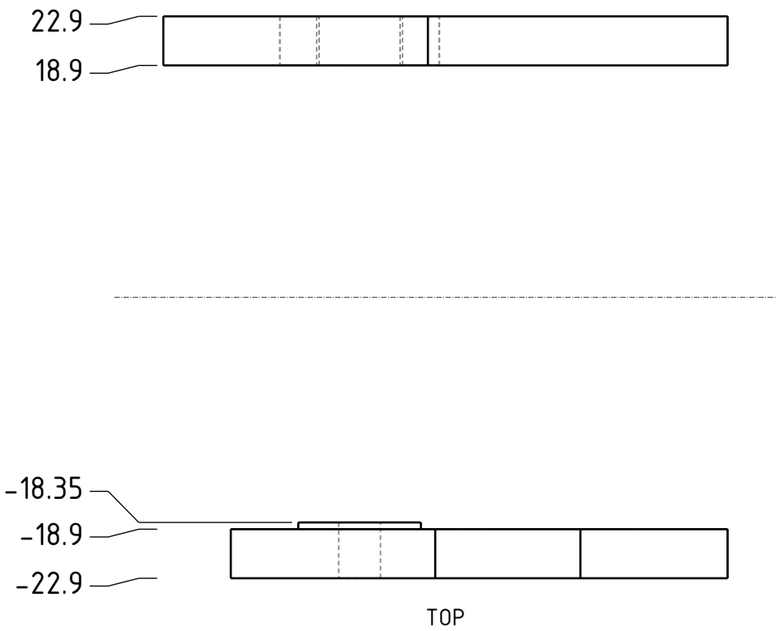
FRONT



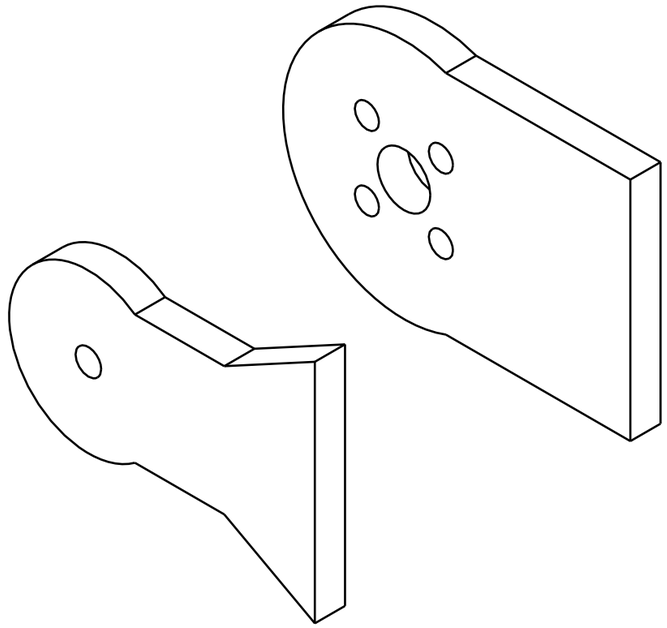
RIGHT

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Servo pocket P (cut into parts) (part sheet 2 of 2)	SCALE 2:1
PART NO. TAL-001	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-04
MATERIAL -	SHEET 3 of 20



TOP



ISOMETRIC (NOT TO SCALE)

IDLER NECK POINTS (X, Z), MIRRORED IN Z

PT	X	Z
N1	0	±10.5
N2	6	±8.5
N3	0.6R = 18	±8.5
N4	R = 30	±H = ±15

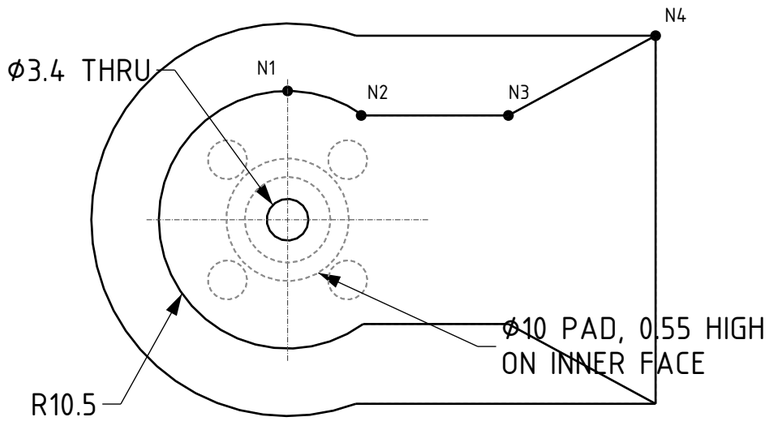
CLEVIS (every pitch joint)

Drawn for the arm links: reach R = 30, half-height H = 15. The wrist uses R = 24, H = 12.

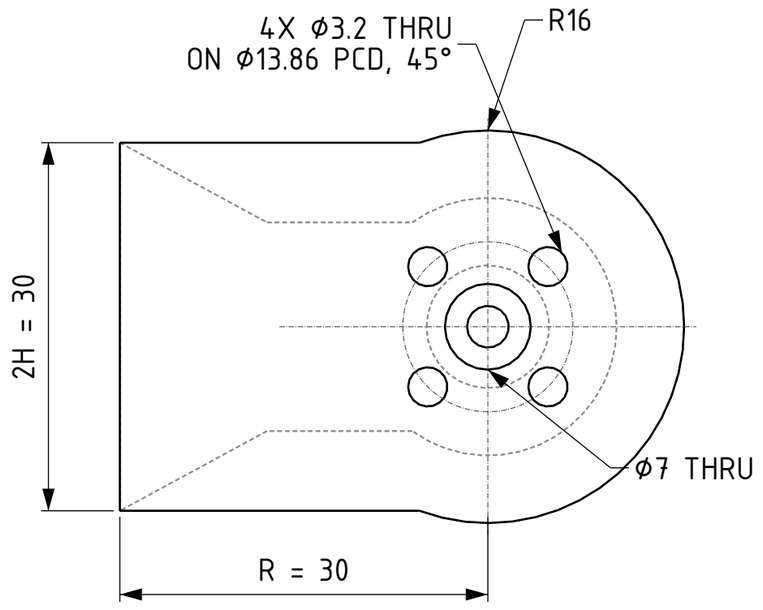
1. Output plate (+Y), y 18.9 to 22.9: circle R16 at the origin + rectangle x 0 to R, z ±H.
2. Output plate holes, through: Ø7 centre (horn spline screw) + 4X Ø3.2 on Ø13.86 PCD at 45°.
3. Idler plate (-Y), y -22.9 to -18.9: circle R10.5 + neck polygon N1 to N4 (mirrored). The neck clears the parent servo's bus plugs.
4. Idler pad Ø10 on the inner face, y -18.9 to -18.35; centre hole Ø3.4 through.
5. Clevis gap in the link body: remove a cylinder R21 about Y between y ±18.9, and everything at x < 0 between those planes (dimensioned on each link's section).

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.



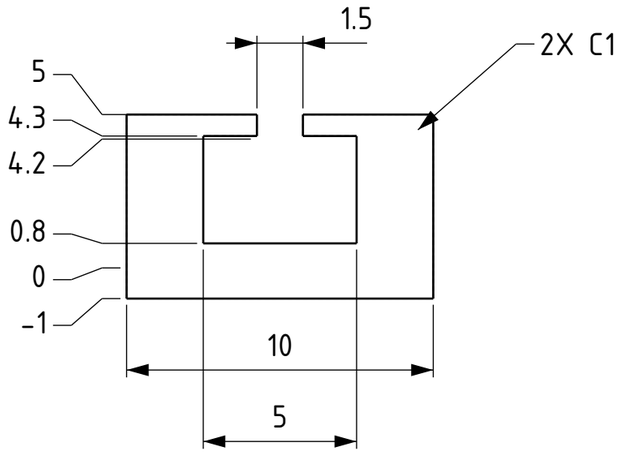
FRONT (IDLER PLATE, -Y)



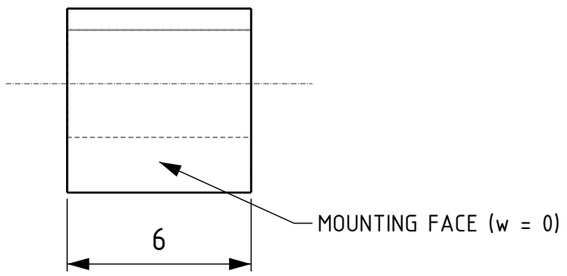
REAR (OUTPUT PLATE, +Y)

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Common features: clevis (part sheet 1 of 2)	SCALE 2:1
PART NO. TAL-002	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-04
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 4 of 20



CABLE CLIP: END VIEW



CABLE CLIP: SIDE VIEW

SEAT S (FROM THE SERVO MID-PLANE)

PART	S
arm shells, gripper	17.3
shoulder	18.5
base	18.4
wrist	18.0

COMMON FEATURES

CABLE CLIP, one bus cable. Local u along the cable, v across, w out of the mounting face.

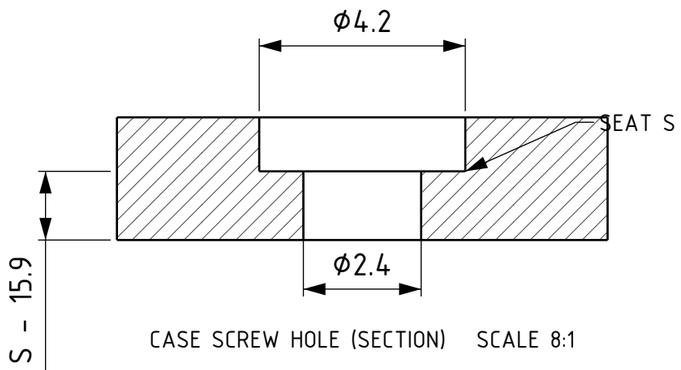
1. Block u ±3, v ±5, w -1 to 5 (sunk 1 into the face).
2. Tunnel through along u: v ±2.5, w 0.8 to 4.3. Slit v ±0.75 from w 4.2 out through the top.
3. Chamfer C1 on the two outer long edges.

CASE SCREW HOLE (M2 self-tap into the servo case): ⌀2.4 from the servo face to the seat S, then ⌀4.2 counterbore from S outward through the part.

HEX WINDOW H(xa, xb, zc, k): hexagon (xa, zc) (xa+k, zc-k) (xb-k, zc-k) (xb, zc) (xb-k, zc+k) (xa+k, zc+k), cut through the part along Y. Drawn for H(0, 40, 0, 10).

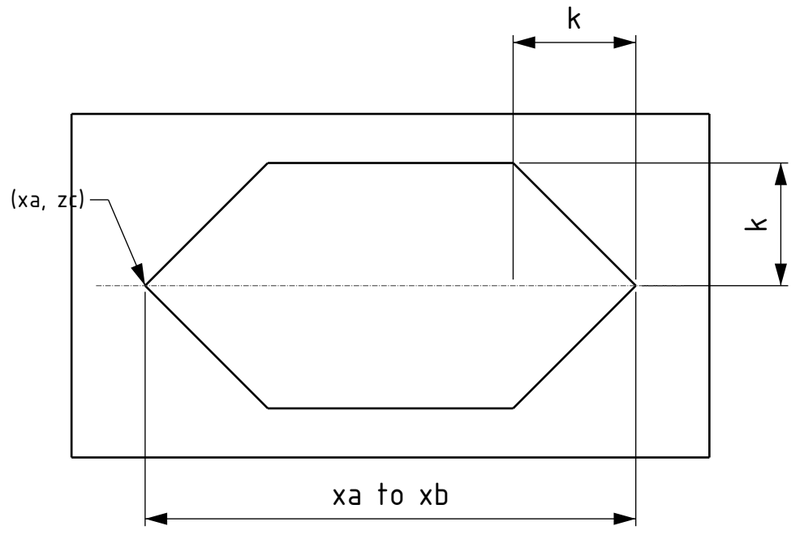
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.



CASE SCREW HOLE (SECTION) SCALE 8:1

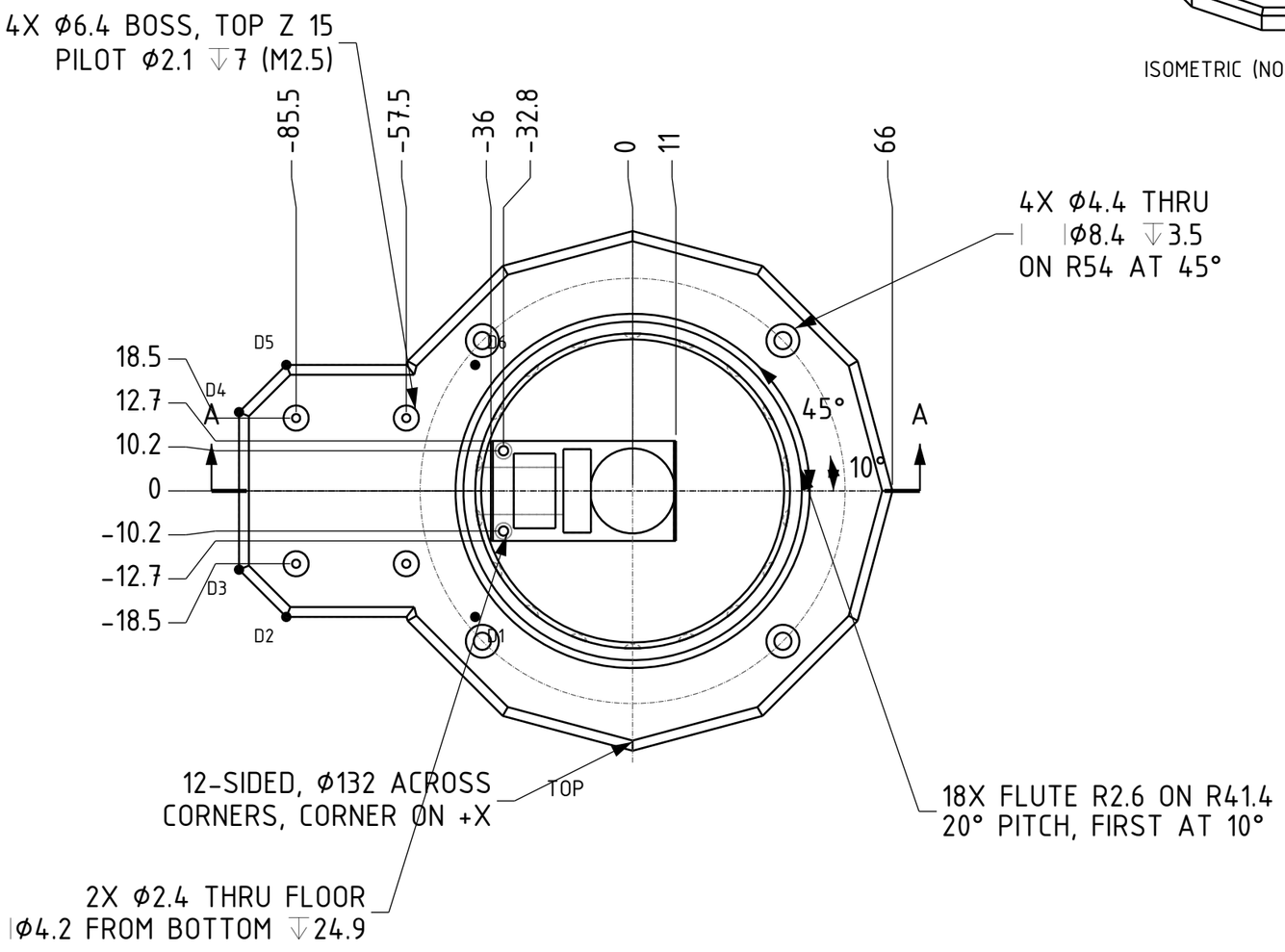
SERVO FACE (BOTTOM) IS 15.9
FROM THE SERVO MID-PLANE

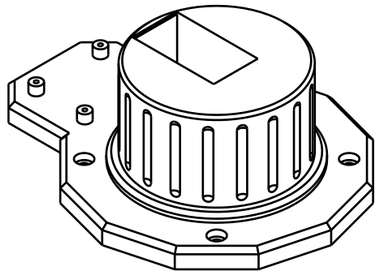


HEX WINDOW H(xa, xb, zc, k) SCALE 2:1

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Common features: clip, screw hole, window (part sheet 2 of 2)	SCALE 5:1
PART NO. TAL-002	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-04
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 5 of 20





ISOMETRIC (NOT TO SCALE)

BOARD DECK POINTS (X, Y)

PT	X	Y
D1	-40	-32
D2	-88	-32
D3	-100	-20
D4	-100	20
D5	-88	32
D6	-40	32

FEATURES (in order)

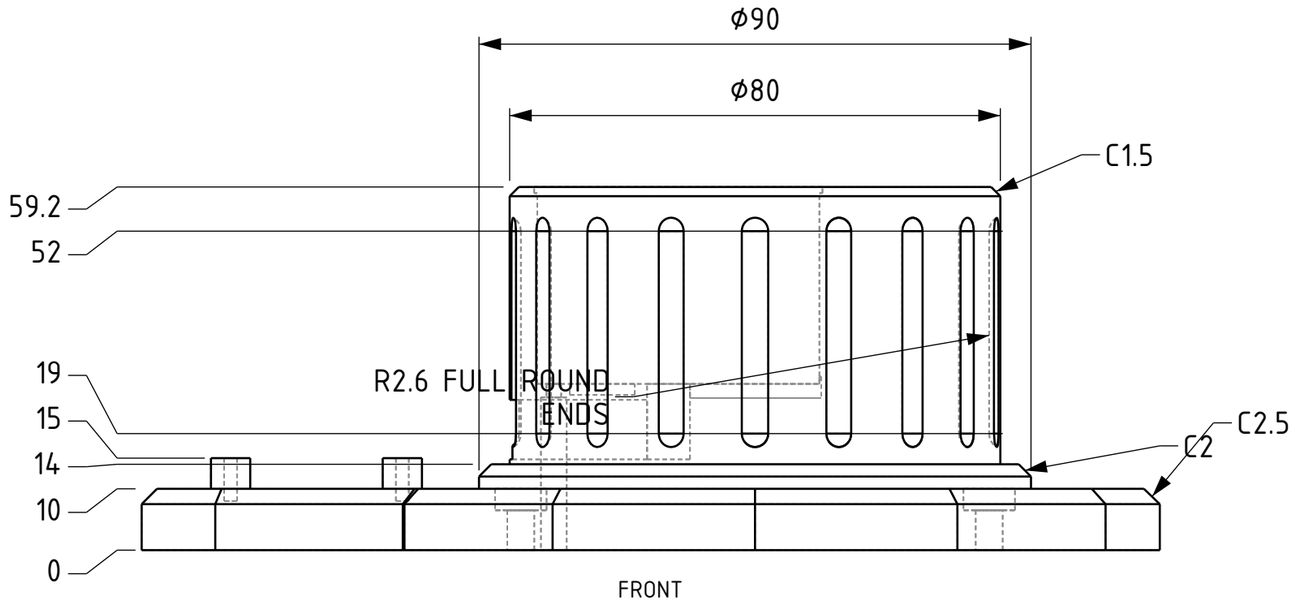
1. Plinth: regular 12-gon, ⌀132 across corners (a corner on +X), z 0 to 10.
2. Board deck: polygon D1 to D6, z 0 to 10; union with the plinth.
3. Chamfer C2.5 on all top outer edges (z 10).
4. Mounting holes: 4X ⌀4.4 through on R54 at 45° + 90°n; counterbore ⌀8.4 from z 6.5 up.
5. Collar ⌀90, z 10 to 14, C2 on its top edge.
6. Column ⌀80, z 10 to 59.2, C1.5 on its top edge.
7. Board standoffs: 4X ⌀6.4, z 9 to 15, at x -57.5 / -85.5, y ±18.5; pilot ⌀2.1 from z 8 to 16.
8. Flutes: 18X vertical ⌀5.2 on R41.4 at 10° + 20°n, axis z 19 to 52, spherical ends; cut.
9. Pocket P (sheet TAL-001) with its servo frame origin P0 at (0, 0, 43.3): servo +Z (shaft) along +Z, servo +X along +X, servo +Y along +Y.
10. Top opening: cut box x -36 to 11, y ±12.7, from z 58.2 up.
11. Cable tunnel (plug bay to board deck): cut box x -60 to -13, y ±6, z 14.8 to 24.5.
12. Case screw holes from below (TAL-002, seat 18.4) at x -32.8, y ±10.2: ⌀2.4 z 24.9 to 27.5, ⌀4.2 counterbore from the bottom to z 24.9.

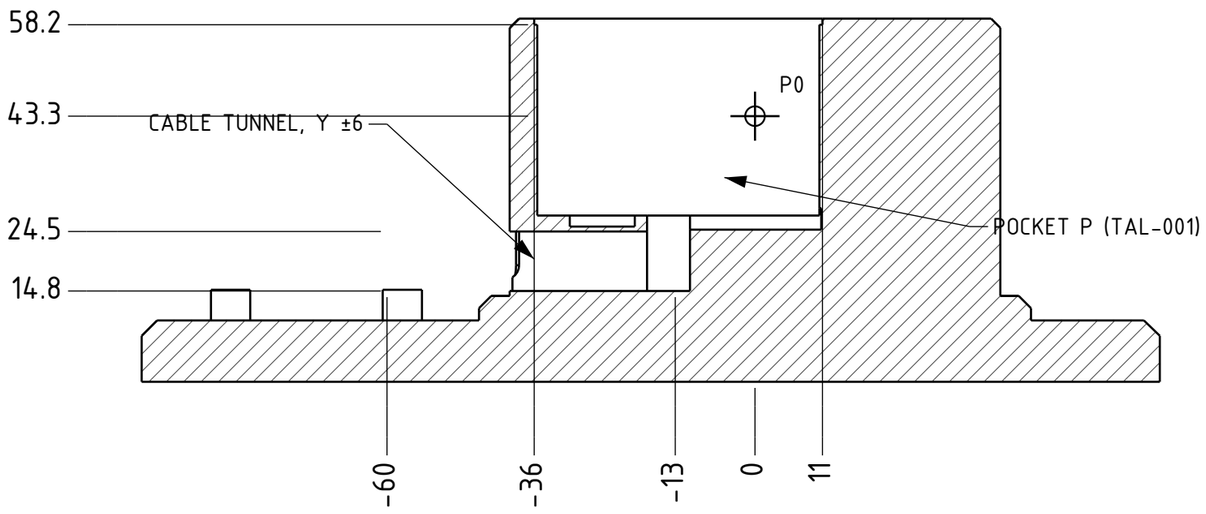
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- The pan servo drops in from the top; its plugs hang into the bay under it.

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps	
PROJECT TALOS 5-axis arm	EST. MASS 155 g
PART Base (part sheet 1 of 2)	SCALE 2:3
PART NO. TAL-101	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-04
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 6 of 20





SECTION A-A

SECTION A-A

Section on the XZ plane (cutting line on sheet 1), looking toward +Y.

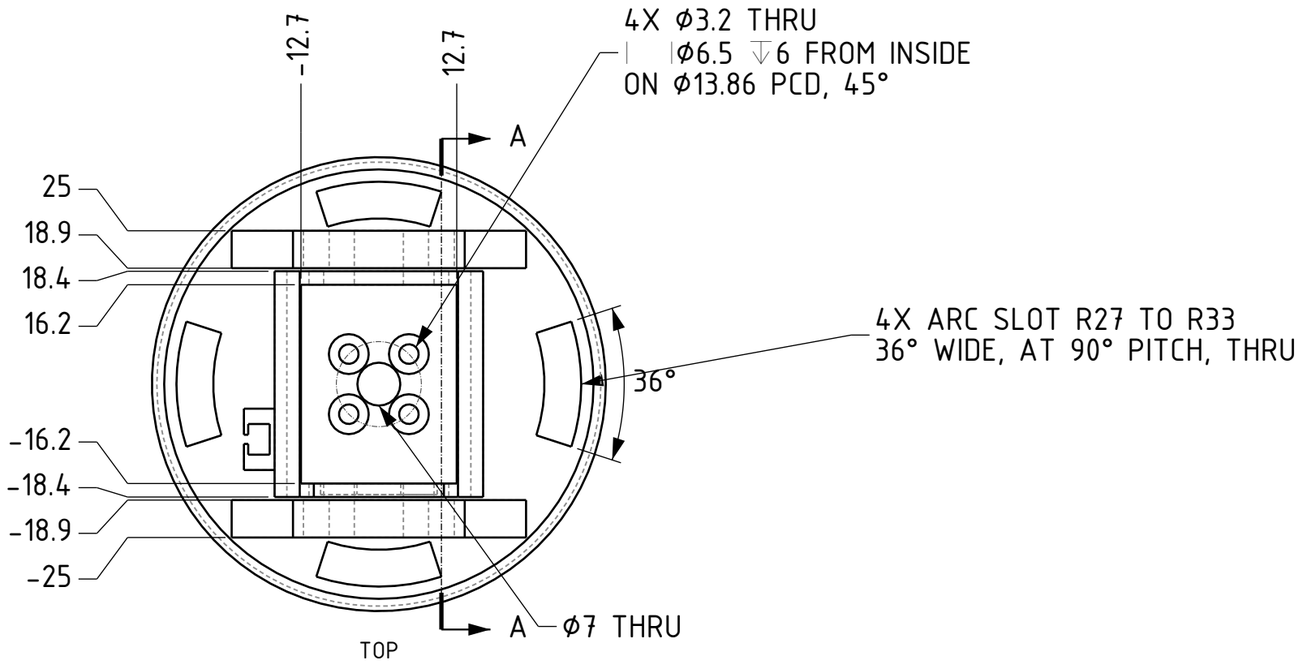
P0 = pocket P servo frame origin; servo axes along the part axes.

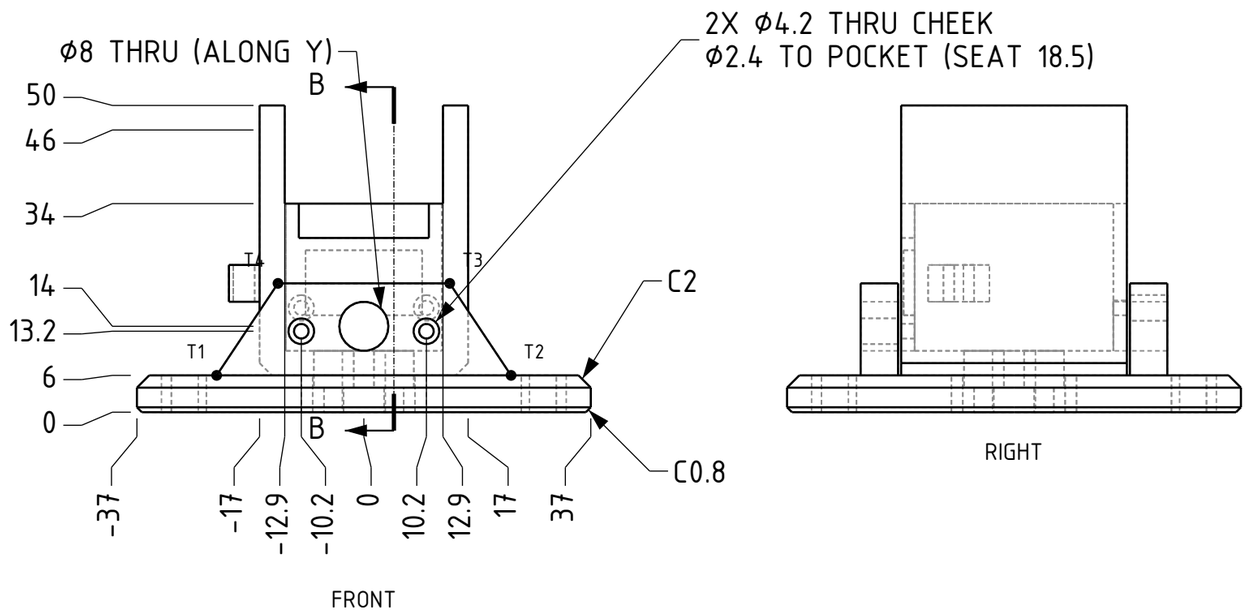
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Base (part sheet 2 of 2)	SCALE 1:1
PART NO. TAL-101	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-04
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 7 of 20





CHEEK PROFILE (X, Z)

PT	X	Z
T1	-24	6
T2	24	6
T3	14	21
T4	-14	21

FEATURES (in order)

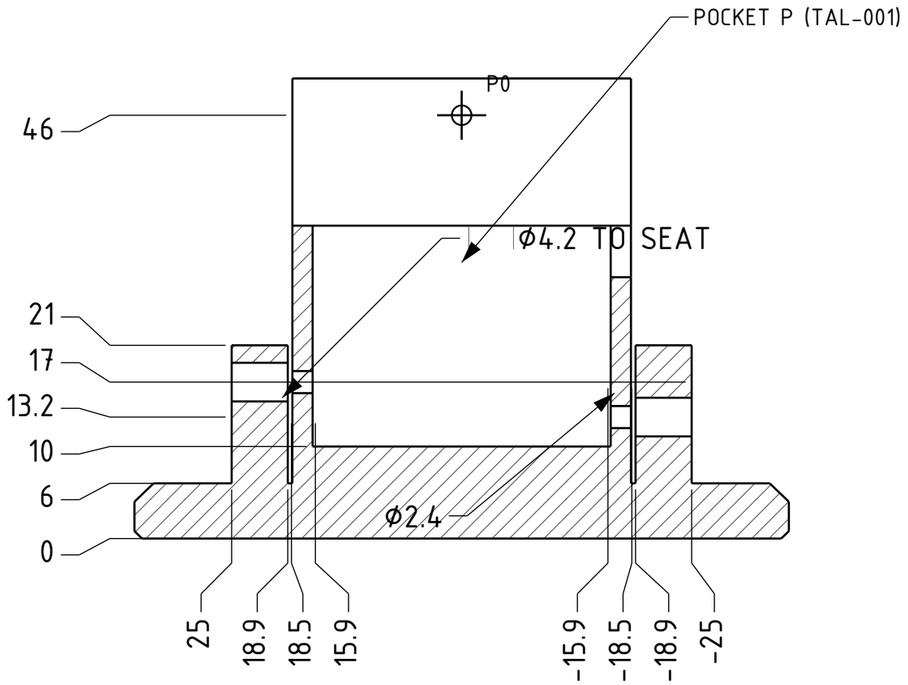
1. Turntable Ø74, z 0 to 6; C2 top edge, C0.8 bottom edge.
2. Cradle: box x ±17, y ±18.4, z 5 to 34.
3. Posts: boxes x -17 to -12.9 and 12.9 to 17, y ±18.4, z 30 to 50.
4. Cheeks: profile T1 to T4, y 18.9 to 25 and -25 to -18.9.
5. Cable clip (TAL-002) on the -X face: centre (-17, -9, 21), cable along Z.
6. Pocket P (sheet TAL-001) with its servo frame origin P0 at (0, 0, 46): servo +Z (shaft) along +Y, servo +X along +Z (body hangs down), servo +Y along +X.
7. Servo opening: cut box x ±12.7, y ±16.2, from z 10 up.
8. Cable hole Ø8 along Y, through all, at x 0, z 14.
9. Pan horn: Ø7 centre through. 4X Ø3.2 through on Ø13.86 PCD at 45°, counterbore Ø6.5 from the floor (z 10) down to z 4 (M3 × 6 from inside the cradle).
10. Turntable windows: 4X arc slot R27 to R33, ±18° about 0°, 90°, 180°, 270°, through.
11. Case screw holes (TAL-002, seat 18.5) through the cheeks: +Y cheek at x ±10.2, z 17; -Y cheek at x ±10.2, z 13.2.

NOTES

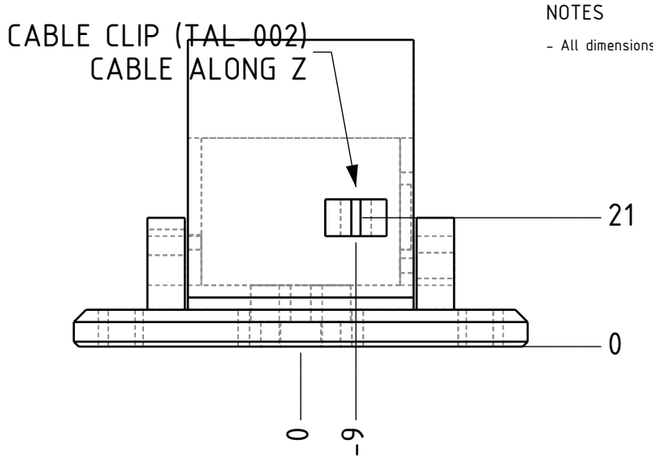
- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps	
PROJECT TALOS 5-axis arm	EST. MASS 39 g
PART Shoulder (turntable + cradle) (part sheet 1 of 2)	SCALE 1:1
PART NO. TAL-102	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-04
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 8 of 20



SECTION A-A



LEFT SCALE 1:1

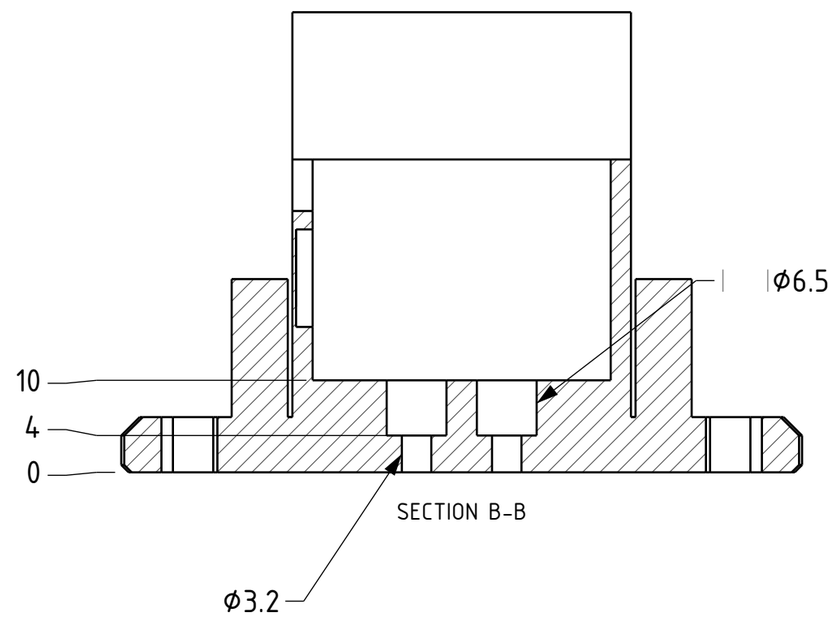
SECTIONS

A-A: plane x 10.2 through one hole of each case screw row (cutting line on sheet 1, top view), looking toward +X. The other holes mirror at x -10.2.

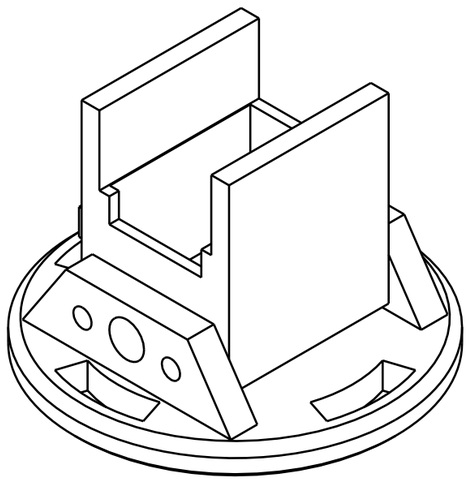
B-B: plane x 4.9 through two pan horn screw holes (cutting line on sheet 1, front view).

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.



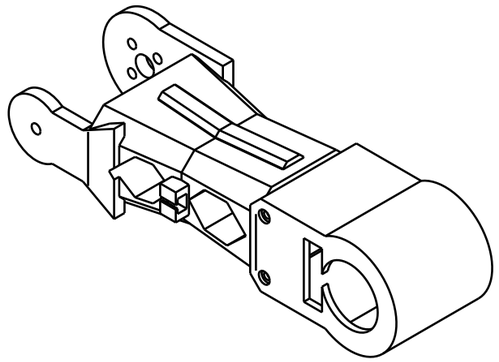
SECTION B-B



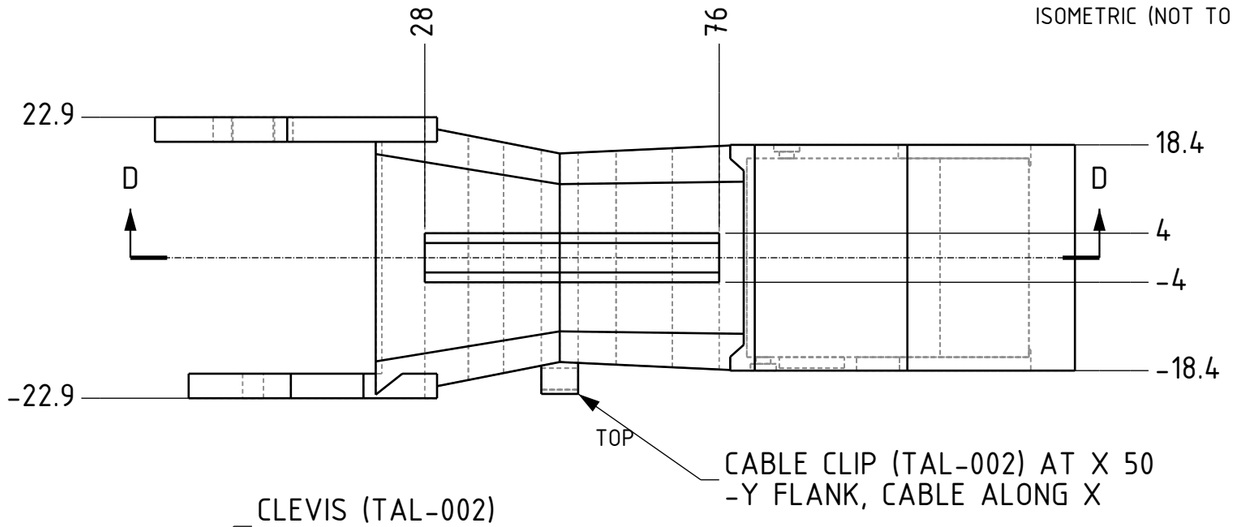
ISOMETRIC (NOT TO SCALE)

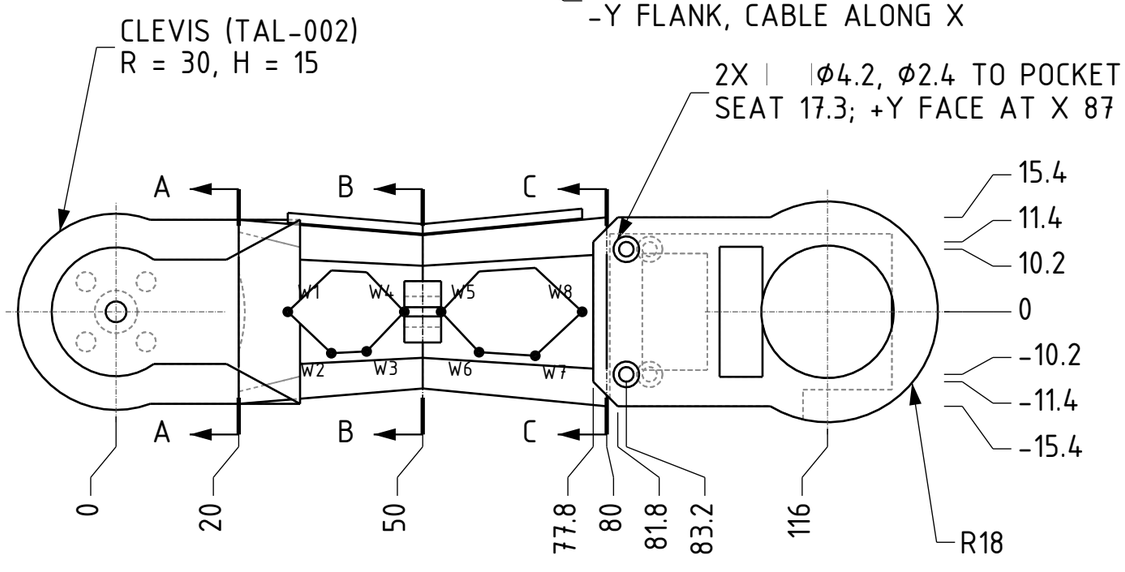
THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Shoulder (turntable + cradle) (part sheet 2 of 2)	SCALE 3:2
PART NO. TAL-102	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-04
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 9 of 20



ISOMETRIC (NOT TO SCALE)





FRONT

WINDOW POINTS (X, Z), THROUGH Y

PT	X	Z
W1	28	0
W2	35.1	±6.75
W3	40.85	±6.46
W4	47	0
W5	53	0
W6	59.19	±6.58
W7	68.35	±7.16
W8	76	0

FEATURES (in order)

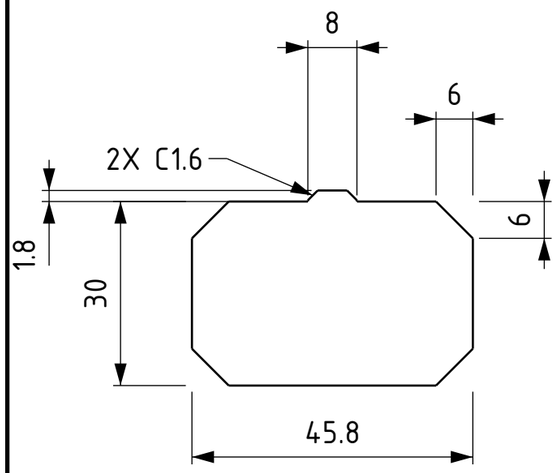
1. Clevis (TAL-002), R = 30, H = 15.
2. Beam: ruled loft of the octagon profiles A, B, C (sheet 2) at x 20, 50, 80.
3. Servo shell: XZ polygon (81.8, ±15.4) (116, ±15.4) (77.8, ±11.4), extruded y ±18.4; plus a cylinder R18 about Y at x 116.
4. Spine: ruled loft of the ridge profiles A, B, C, trimmed to x 28 to 76.
5. Windows: polygons W1 to W4 and W5 to W8 (mirrored in Z), cut through Y.
6. Cable clip (TAL-002) on the -Y flank at x 50, cable along X.
7. Clevis gap cut (TAL-002): R21 about Y between y ±18.9, plus x < 0 between those planes.
8. Pocket P (sheet TAL-001) with its servo frame origin P0 at (116, 0, 0): servo +Z (shaft) along +Y, servo +X along +X, servo +Y along -Z.
9. Servo opening (drops in from below): cut box x 80.5 to 112, y ±16.2, z below -12.
10. Case screw holes (TAL-002, seat 17.3): +Y face at x 87, z ±10.2; -Y face at x 83.2, z ±10.2.

NOTES

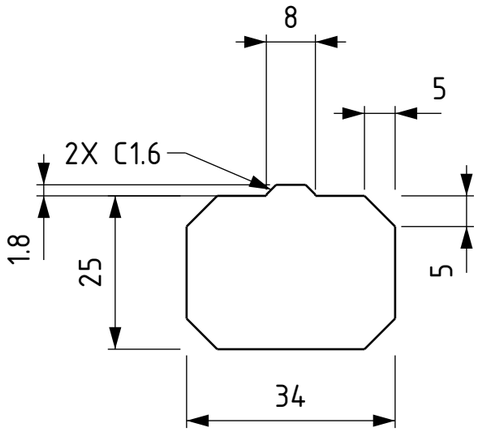
- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- The elbow servo's bus plugs come out through the -Y shell wall.

THIRD ANGLE PROJECTION

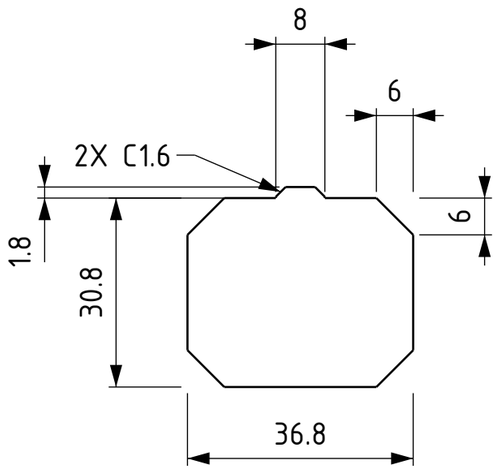
WORK SET A: base, shoulder, upper arm, hub caps	
PROJECT TALOS 5-axis arm	EST. MASS 51 g
PART Upper arm (part sheet 1 of 2)	SCALE 1:1
PART NO. TAL-103	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-04
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 10 of 20



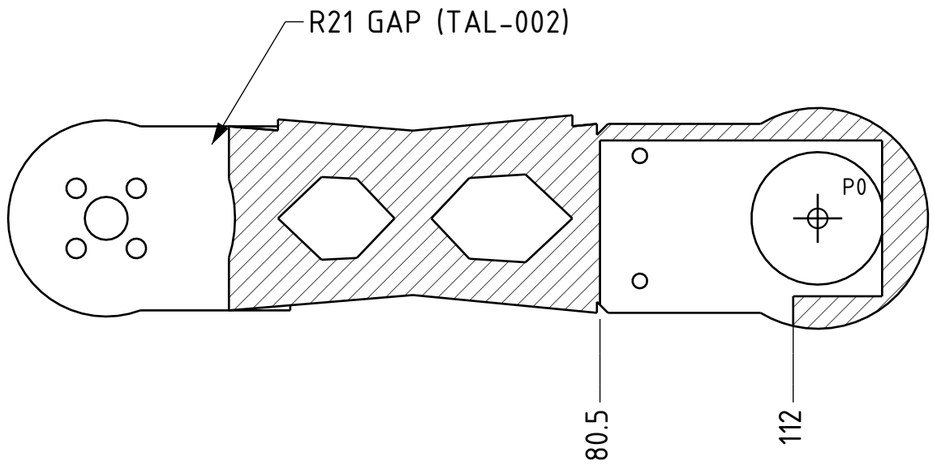
PROFILE A–A (X 20)



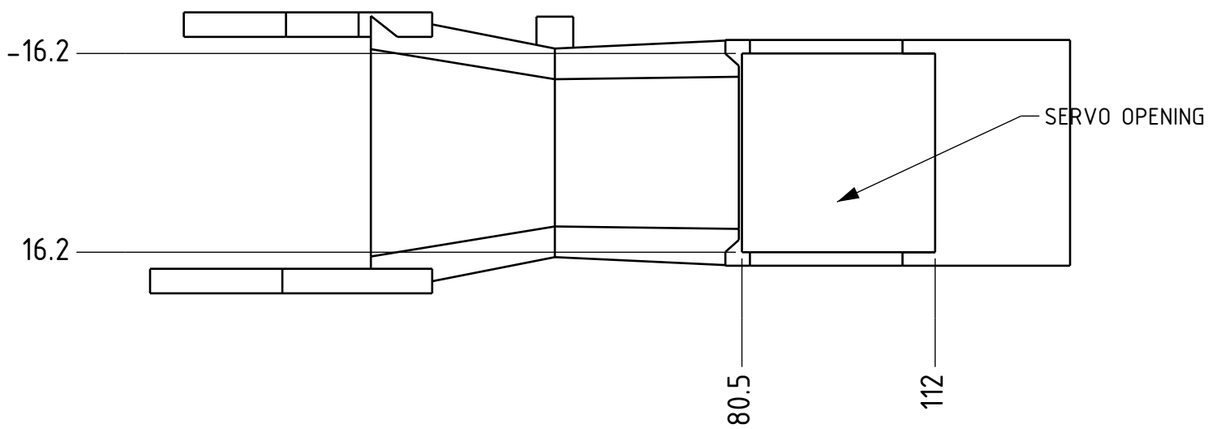
PROFILE B–B (X 50)



PROFILE C–C (X 80)



SECTION D–D



BOTTOM

PROFILES AND SECTION

Profiles A, B, C: beam octagon (width × height, corners chamfered) with the spine ridge on top, in the YZ plane at the loft stations; cutting lines on sheet 1.

Ridge: 8 wide, 4.8 across the top, 1.8 above the beam, 45° bevels; its base runs 3 into the beam.

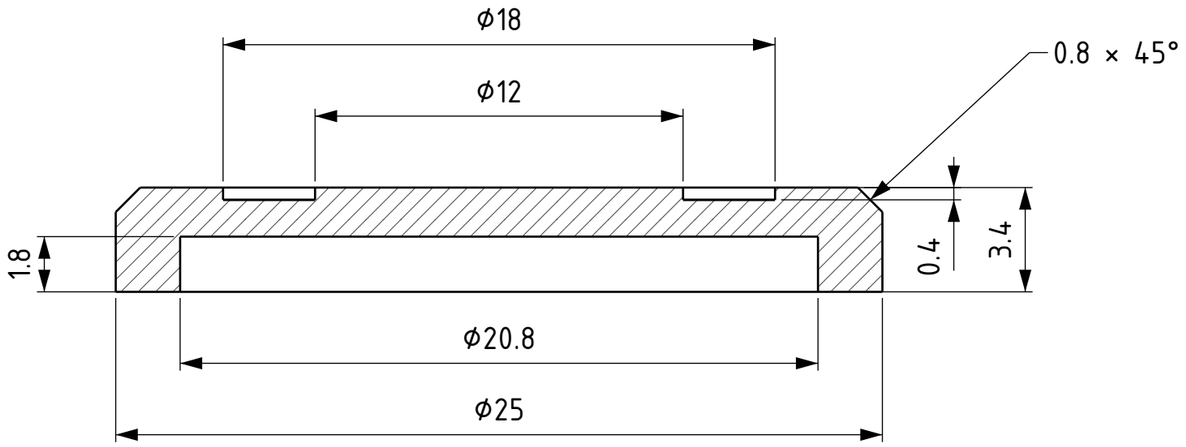
Section D–D: XZ plane, looking toward +Y. P0 = pocket P servo frame origin.

NOTES

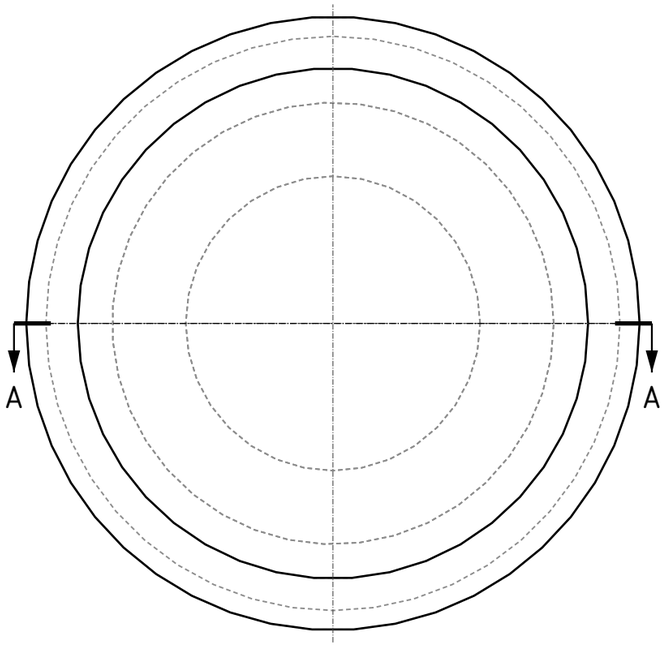
- All dimensions mm. Third-angle projection. Do not scale the drawing.

THIRD ANGLE PROJECTION

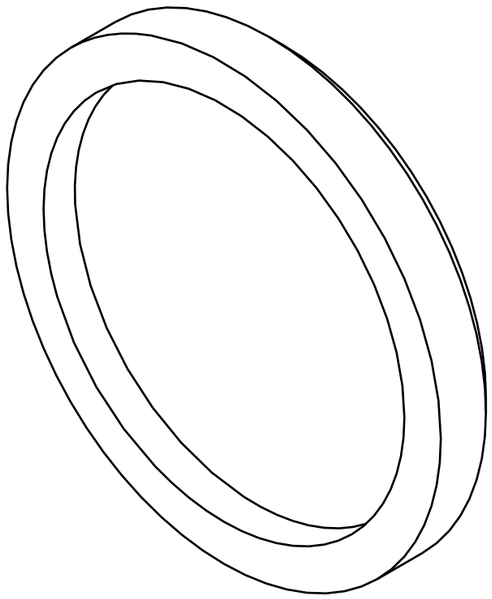
WORK SET A: base, shoulder, upper arm, hub caps	
PROJECT TALOS 5-axis arm	EST. MASS –
PART Upper arm (part sheet 2 of 2)	SCALE 1:1
PART NO. TAL-103	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-04
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 11 of 20



SECTION A–A SCALE 5:1



FRONT (INNER FACE)



ISOMETRIC (NOT TO SCALE)

FEATURES (in order)

HUB CAP, LARGE (output side, covers 4 horn screws):

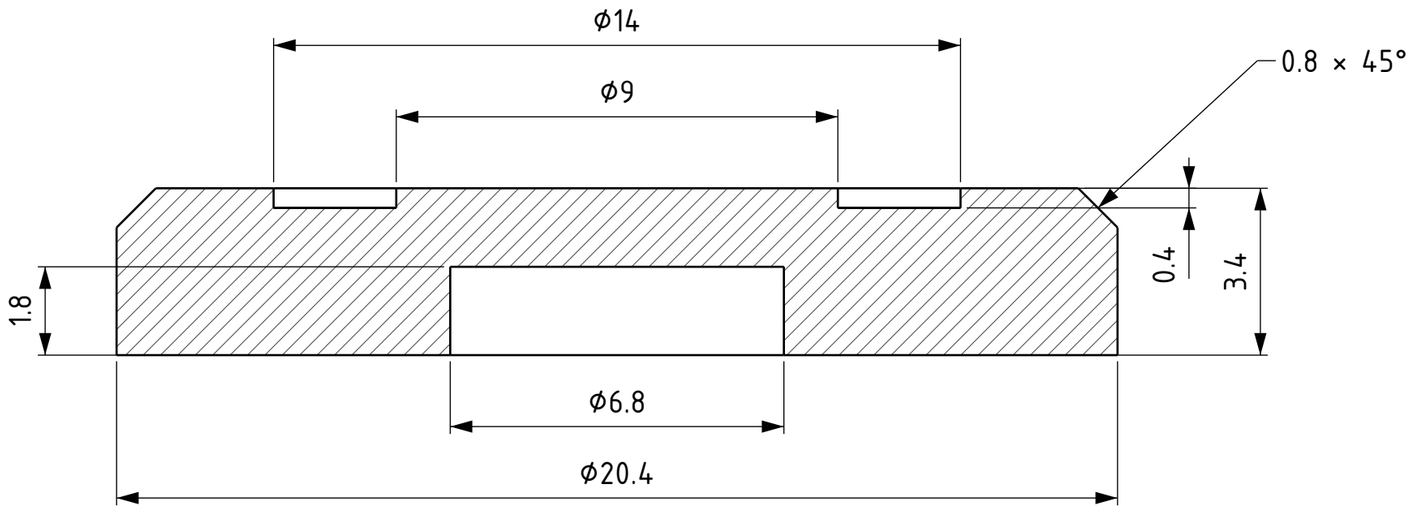
1. Disc ϕ25, 3.4 thick (y 0 to 3.4); chamfer 0.8 × 45° on the outer edge of the outer face.
2. Recess ϕ20.8, 1.8 deep from the inner face (clears M3 button heads, 1.65 high).
3. Face groove: annulus ϕ12 to ϕ18, 0.4 deep from the outer face.

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- Press-fit or glue onto the clevis plate over the screw heads. Print outer face down.

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps	
PROJECT TALOS 5-axis arm	EST. MASS 1 g
PART Hub cap, large	SCALE 4:1
PART NO. TAL-108	UNITS / REV mm / A
QTY / ARM 3	DATE 2026-11-04
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 19 of 20



SECTION A–A SCALE 8:1

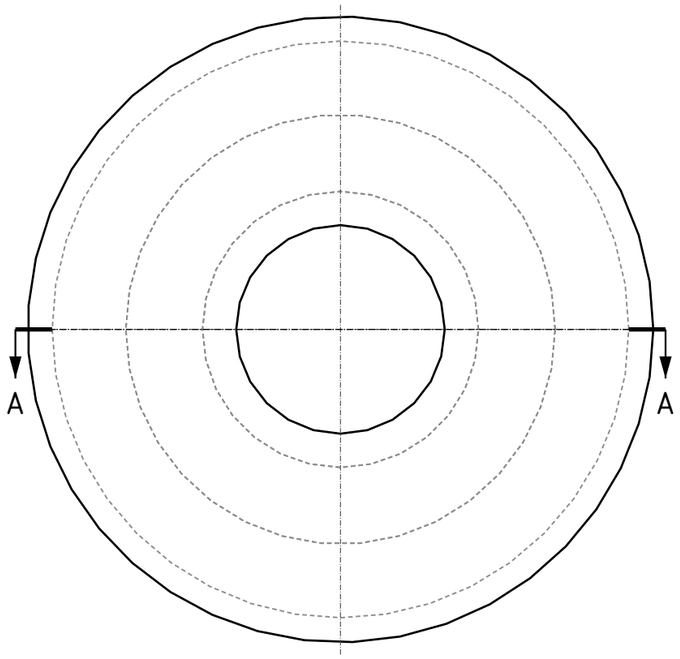
FEATURES (in order)

HUB CAP, SMALL (idler side, covers 1 centre screw):

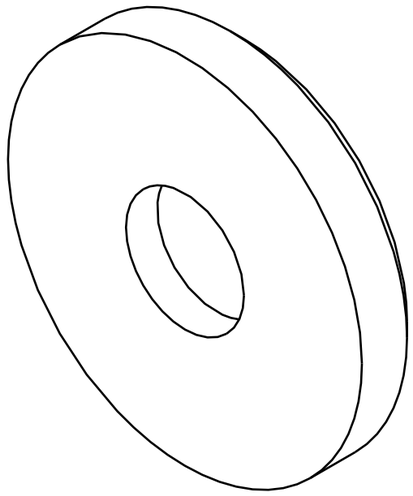
1. Disc ϕ20.4, 3.4 thick (y 0 to 3.4); chamfer 0.8 × 45° on the outer edge of the outer face.
2. Recess ϕ6.8, 1.8 deep from the inner face (clears M3 button heads, 1.65 high).
3. Face groove: annulus ϕ9 to ϕ14, 0.4 deep from the outer face.

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- Press-fit or glue onto the clevis plate over the screw heads. Print outer face down.



FRONT (INNER FACE)



ISOMETRIC (NOT TO SCALE)

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps	
PROJECT TALOS 5-axis arm	EST. MASS 1 g
PART Hub cap, small	SCALE 5:1
PART NO. TAL-109	UNITS / REV mm / A
QTY / ARM 3	DATE 2026-11-04
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 20 of 20